

PROGRAM TO SEARCH AN ELEMENT IN ARRAY USING BINARY SEARCH

AIM:

To write array program using binary concepts

ALGORITHM:

STEP 1: start

STEP 2: declare no of elements (n) elements to search (key)

STEP 3: initialization array with n(arr[n])

STEP 4: i=0

STEP 5: get arr[i]

STEP 6: i++

STEP 7: if i<n goto step 5,else continue

STEP 8: read element to search(key)

STEP 9: result = binary search(arr,n,key)

STEP 10: if (result!=1)

STEP 11: print element found at position(result)else
print element not found

STEP 12: stop

ALGORITHM FOR BINARY SEARCH:

STEP 1: int low=0, high=n-1

STEP 2: if low(<=high)

STEP 3: mid=(low+high)/2

STEP 4: if (arr[mid]==key)

return mid

STEP 5: else if (arr[mid]<key)

low=mid+1

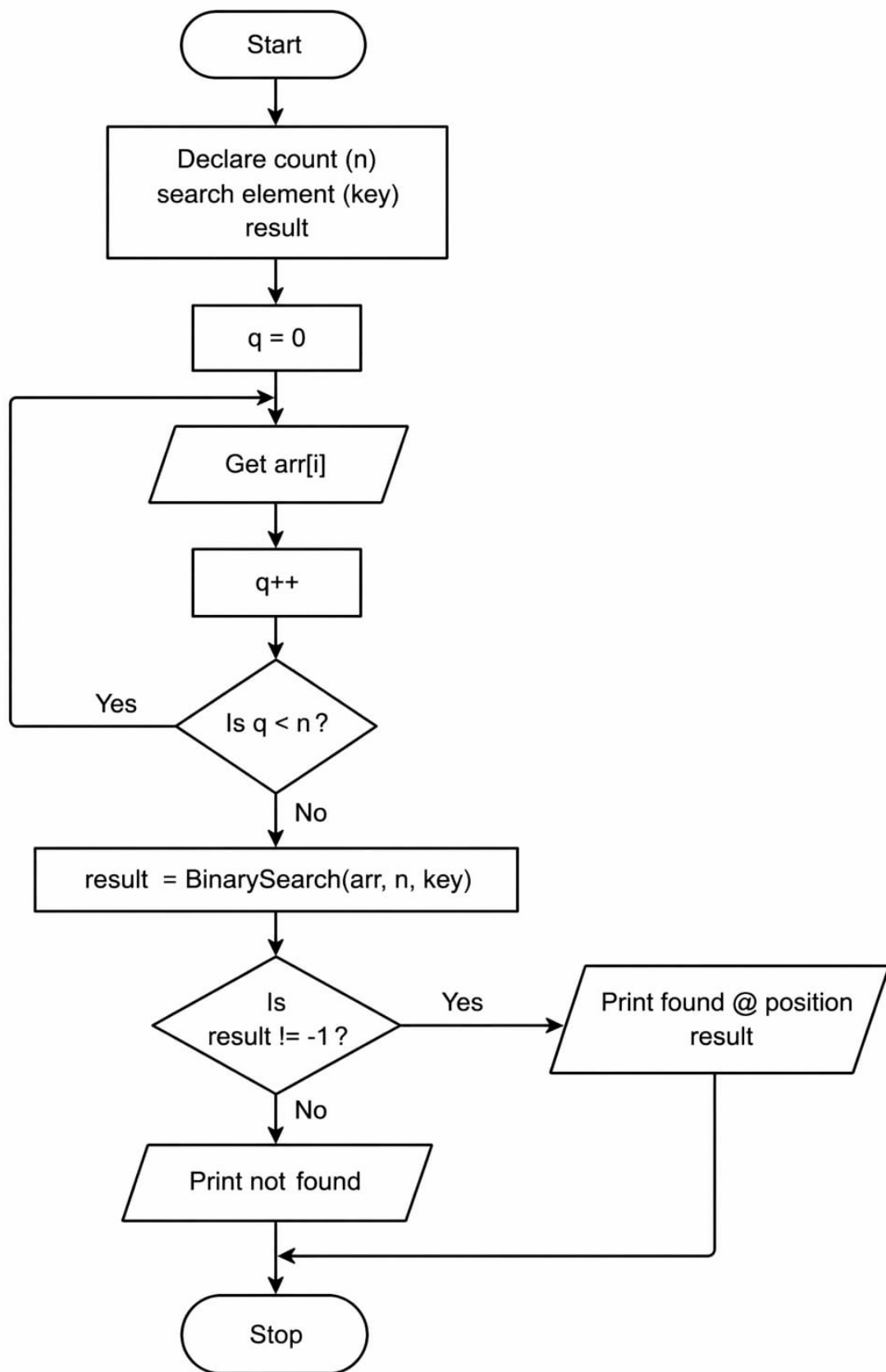
else

high+mid-1

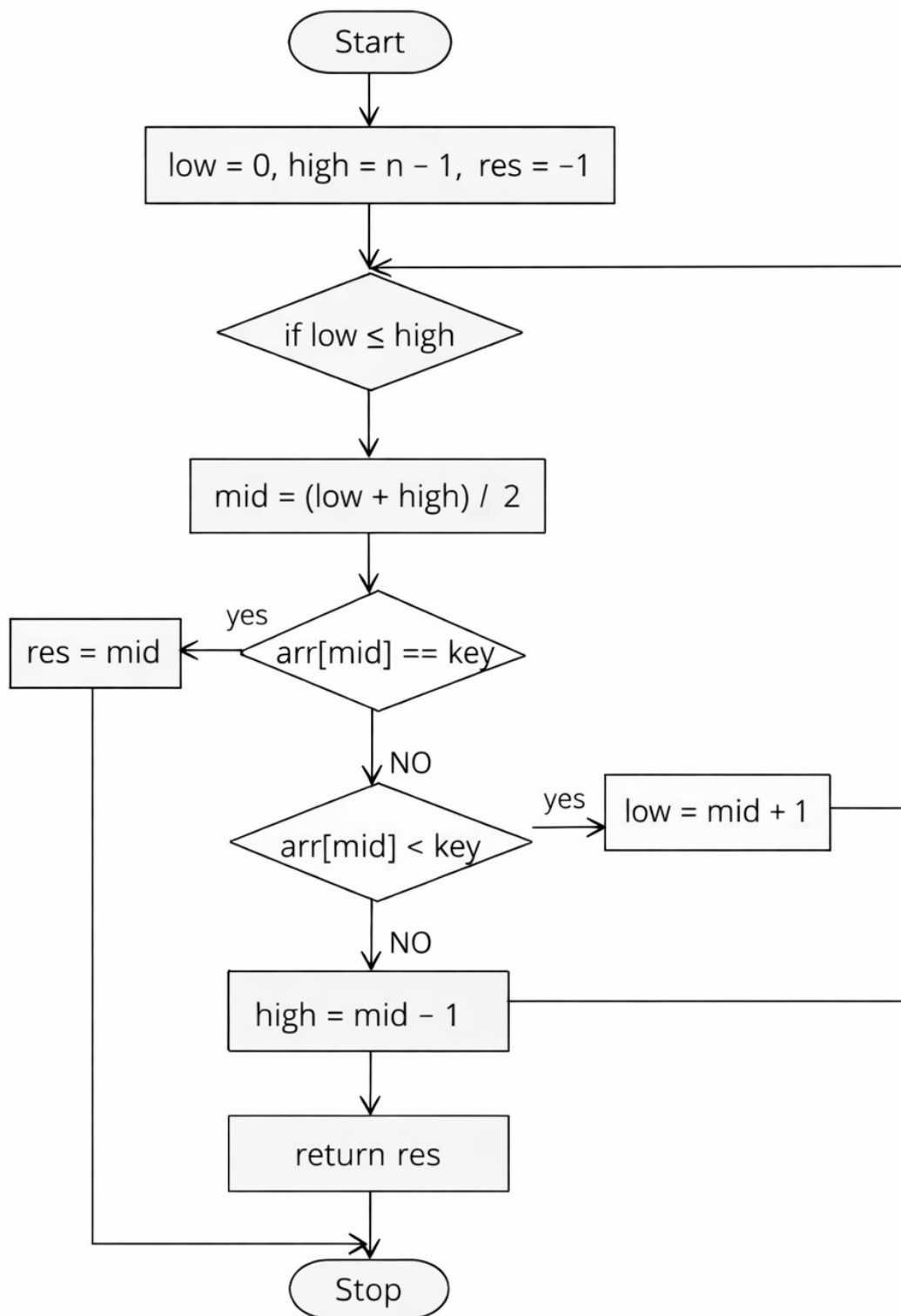
STEP 6: goto step 2

STEP 7: return-1

FLOWCHART :



ALGORITHM FOR BINARY SEARCH :



CODING :

```
#include<stdio.h>
int binarysearch(int arr[],int n, int key)
{
int low=0,high=n-1,mid;
while(low<=high)
{
mid=(low+high)/2;
if (arr[mid]==key)
return mid;
//element found
else if(arr[mid]<key)
low=mid+1;
//search right half
else
high=mid-1;
//search left half
}
return 1;
//element not found
}
int main()
{
int n,key,result;
printf("enter number of element:");
scanf("%d",&n);
int arr[n];
printf("enter %d sorted element:\n",n);
int i;
for(i=0;i<n;i++){
scanf("%d",&arr[i]);
```

```
}  
printf("enter element to search:");  
scanf("%d",&key);  
result = binarysearch(arr, n, key);  
if(result!=-1)  
printf("element found at index %d \n",result);  
else  
printf("element not found \n");  
return 0;  
}
```

OUTPUT :

```
enter no of element   : 5  
enter 5 sorted element  
11  
12  
13  
14  
15  
enter element to search : 14  
enter found at index : 3
```

RESULT :

Thus the program compiled and executed successfully with verified output